

LibTorch : Day 2

Tensors - The Building Blocks Of Everything

What is a Tensor?

→ Imagine a number line (1 Dimensional), then a spreadsheet table (2 Dimensional), then a cube of numbers (3D).

A 'tensor' is all of it, and MORE dimensions.

↳ A 2D photo is a tensor

↳ A color photo is a 3D tensor

↳ A batch of color photo is a 4D Tensor

★ Tensors are just number grids with any number of dimensions.

Tensor Dimensions

0D Scalar - A single no.

1D Vector - A list of nos.

2D Matrix - A table/spreadsheet

3D Cube - Stack of matrices

4D Hypercube - Batch of 3D tensors

Example

`torch::tensor(42.0)`

`torch::zeros(5)`

`torch::ones({3,4})`

`torch::rand({3,224,224})`

`torch::rand({32,3,224,224})`

Creating Tensors - Every way possible

```
auto zeros = torch::zeros({3, 4}); // All zeros
```

```
auto ones = torch::ones({2, 3}); // All ones
```

```
auto randn = torch::randn({4, 4}); // Random normal  
                                     (mean=0, std=1)
```

```
auto rand = torch::rand({5, 5});
```

* explore 'full', 'eye', 'arange' and 'linspace'

↓
* arange

// — FROM EXISTING DATA —

```
std::vector<float> data = {1.0, 2.0, 3.0, 4.0, 5.0, 6.0};  
auto from-vec = torch::tensor(data);
```

```
float arr[6] = {1, 2, 3, 4, 5, 6};
```

```
auto from-arr = torch::from_blob(arr, {2, 3});
```

from_blob is used to create a tensor that wraps an existing memory buffer without copying the underlying data.

↳ Creates a view of the data.

↓
from_blob does not copy data. The array must stay alive!

// -- SPECIFYING DATA TYPES --

```
auto float64 = torch::zeros({3}, torch::kFloat64);  
// double
```

```
auto int32 = torch::zeros({3}, torch::kInt32);
```

```
auto bool_t = torch::zeros({3}, torch::kBool);
```

// Suppose we do,

```
torch::Tensor t_3D = torch::randn({4, 3, 2});
```

We can do the following operation →

```
std::cout << t_3D.sizes() << std::endl; //  
[4, 3, 2]
```

```
std::cout << t_3D.size(0) << std::endl; //  
[4]
```

```
std::cout << t_3D.numel() << std::endl;
```

↓

```
// 24 (4 * 3 * 2)
```